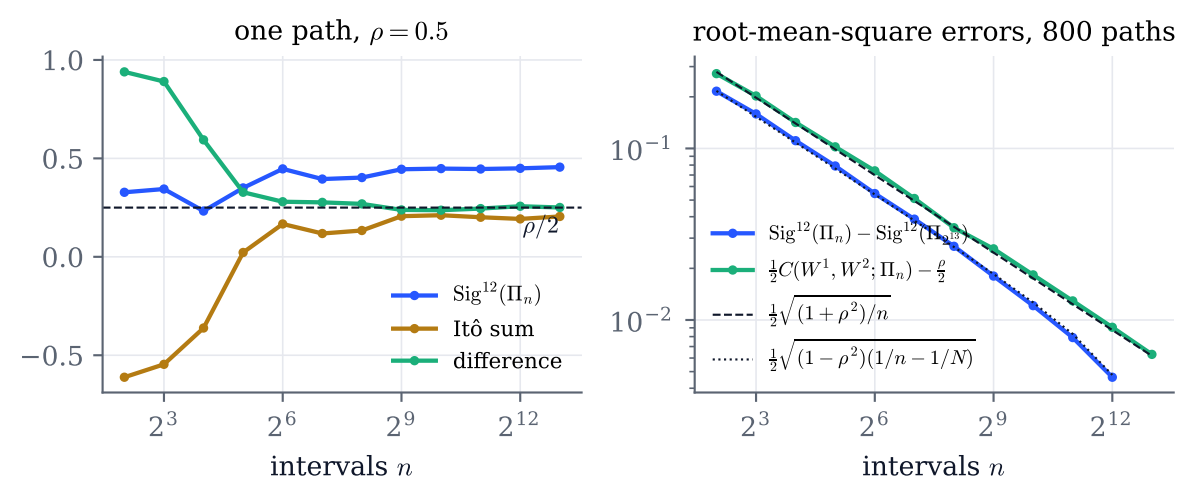
one path, $\rho = 0.5$
1.0
0.5
0.0
−0.5
$\rho/2$
$\mathrm{Sig}^{12}(\Pi_n)$
Itô sum
difference
2^3
2^6
2^9
2^{12}
intervals n
root-mean-square errors, 800 paths
10^{-1}
10^{-2}
$\mathrm{Sig}^{12}(\Pi_n) - \mathrm{Sig}^{12}(\Pi_{2^{18}})$
$\frac{1}{2}C(W^1, W^2; \Pi_n) - \frac{\rho}{2}$
$\frac{1}{2}\sqrt{(1+\rho^2)/n}$
$\frac{1}{2}\sqrt{(1-\rho^2)(1/n - 1/N)}$
2^3
2^6
2^9
2^{12}
intervals n